

NTIC / TLSI

Master 1: SDSI

Module: (Advanced Business Intelligence)

(PROJECT REPORT)

[Electric Vehicle Population Analysis]

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Introduction

This dataset shows the number of Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) registered in U.S. counties, based on data from each state's Department of Licensing (DOL). The analysis aims to provide insights into electric vehicle adoption patterns and support decision-making for various stakeholders.

Functional Study

Decision-Making Need Description

- **Climate-Based Distribution:** Identify EV registration patterns in different climate regions to understand adoption barriers and opportunities.
- **Vehicle Range Analysis:** Determine which electric and hybrid models offer the best range for consumer decision-making.
- **Market Trend Analysis:** Compare popularity between hybrid and fully electric vehicles for investment strategies.
- **Infrastructure Planning:** Identify high-concentration EV areas for targeted charging station deployment.

User Identification

The users of this analysis include:

- Policy makers and government planners
- Investors and EV manufacturers
- customer

Modeling

Conceptual Schema of the Data Warehouse

- **Schema Type:** Star schema optimized for analytical queries
- **Fact Table:** Electric vehicle range metrics
- **Dimension Tables:**
 - Location (County, City, Weather Type)
 - Vehicle (Make, Model, Year, EV Type)

Logical Model

Model Choice Justification

The star schema was selected for:

- Simplified query patterns for business intelligence
- Optimal performance for aggregation operations
- Clear dimensional relationships
- Easy maintenance and scalability

Data Integration (ETL)

Data Sources Used

In this project, I used real data obtained from the following source: <https://catalog.data.gov/dataset/electric-vehicle-population-data>.

Tools and Technologies Used

- **Power BI:** For visualizing the data and creating interactive dashboards.
- **Python:** For data cleaning, transformation, and preparation of the data before loading it into Power BI.

ETL Steps Description

Extraction

The first step in the ETL process was to extract the data from the source. The dataset was downloaded from the website mentioned above in CSV format. This data contains key metrics about the population of electric vehicles in different regions.

Transformation

After downloading the data, Python was used to clean and transform the dataset. The raw data was processed to create two tables:

- **Dimension Table:** Contains descriptive information about the different regions and vehicle types.
- **Fact Table:** Contains the numeric data representing the electric vehicle population in various regions.

This transformation step involved:

- Cleaning missing or invalid data.
- Combining and reshaping the dataset to create meaningful tables.
- Aggregating the data where necessary.

Loading

Once the data was cleaned and transformed, it was loaded into Power BI for further analysis and visualization. Power BI was used to create interactive dashboards that display insights on the electric vehicle population across various regions.

Analysis and Visualization

Electric Vehicle Range by Weather Type

Analysis: The bar chart shows that electric vehicles have a higher range in hot weather conditions compared to cold regions. This may be due to battery performance being more efficient in warmer temperatures, which enhances vehicle mileage.

Electric vs Hybrid Vehicle Distribution

Analysis: From the pie chart, it's evident that electric vehicles make up a larger proportion of the market compared to hybrid vehicles. This suggests growing consumer interest in fully electric solutions over transitional hybrid models.

Top 5 Counties by Vehicle Count

Analysis: The counties of Clark, King, Pierce, Snohomish, and Thurston lead in terms of vehicle count. This could be related to higher population density or urbanization in these areas.

Top 5 Vehicles by Model and Electric Type

Analysis: The bar chart displays the average electric range by model for two types of electric vehicles: Plug-in Hybrid Electric Vehicles (PHEV) and Battery Electric Vehicles (BEV).

For PHEVs, the models shown are WHEGO, RANGE ROVER, CLARITY, and VOLT, with average electric ranges up to 100 miles.

For BEVs, the models include KONA, ROADSTER, MACAN, I-PACE, and MODEL S, with average electric ranges extending up to 300 miles.

Conclusion

In this project, we analyzed data about electric vehicles in the United States. We used Python for data cleaning and transformation, and Power BI for creating visual dashboards. This work helped us understand how to use business intelligence tools to extract useful information from real data.

The project was helpful in applying what we learned in the module, especially about data modeling and visualization. However, some vehicles in the dataset had no data for recent years, which limited the analysis.

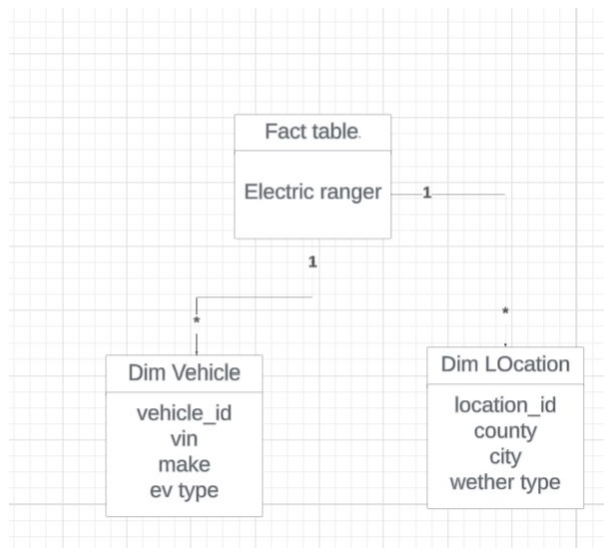


Figure 1: Logical Model

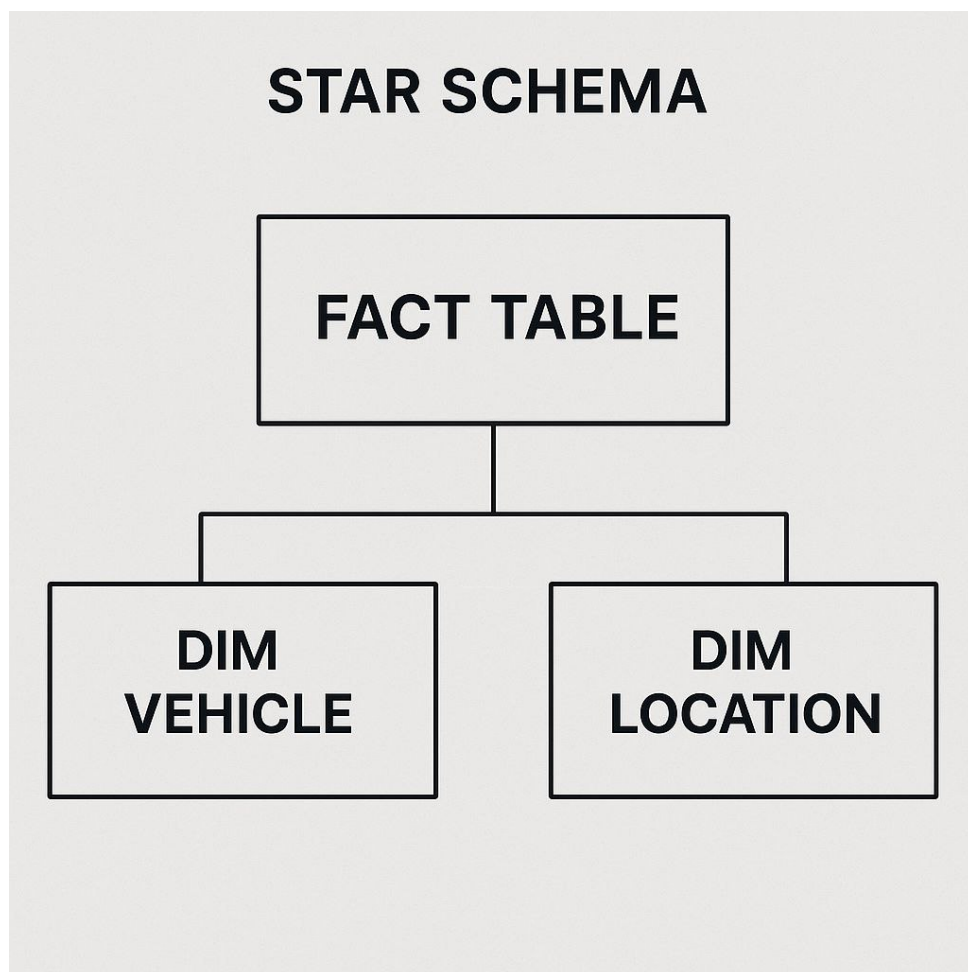


Figure 2: Star Schema Visualization

```

1 import pandas as pd
2
3 df = pd.read_csv("C:/Users/AbdouBmk/Downloads/Electric_Vehicle_Population_Data.csv")
4 df = df.dropna(subset=["Electric Range"])
5
6 dim_vehicle = df[["VIN (1-10)", "Make", "Model", "Model Year", "Electric Vehicle Type"]].drop_duplicates().reset_index(drop=True)
7 dim_vehicle["vehicle_id"] = dim_vehicle.index + 1
8
9 dim_location = df[["County", "City"]].drop_duplicates().reset_index(drop=True)
10 dim_location = dim_location.drop_duplicates(subset=["City"]).reset_index(drop=True)
11 dim_location["location_id"] = dim_location.index + 1
12
13
14 df = df.merge(dim_vehicle, on=["VIN (1-10)", "Make", "Model", "Model Year", "Electric Vehicle Type"], how="left")
15
16 df = df.merge(dim_location, on=["County", "City"], how="left")
17
18 fact_table = df[["vehicle_id", "location_id", "Electric Range"]]
19
20 dim_vehicle.to_csv("dim_vehicle.csv", index=False)
21 dim_location.to_csv("dim_location.csv", index=False)
22 fact_table.to_csv("fact_table.csv", index=False)
23

```

Figure 3: Python code for data transformation and cleaning.

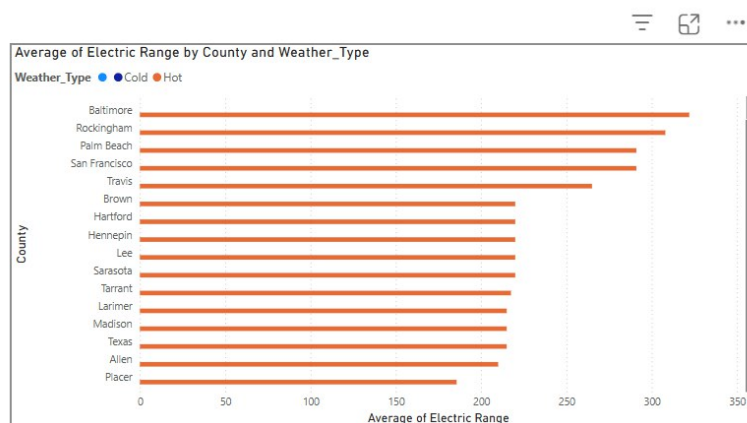


Figure 4: Electric Vehicle Range by Weather Type

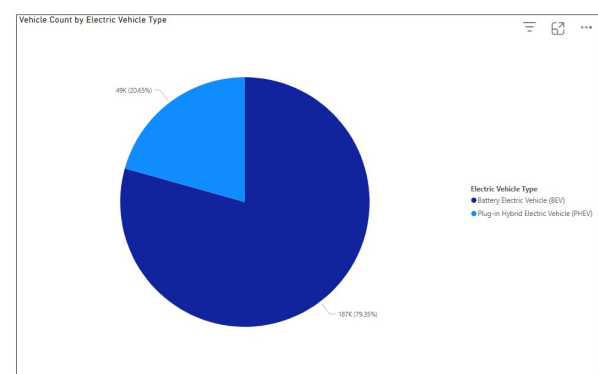


Figure 5: Distribution of Electric vs Hybrid Vehicles

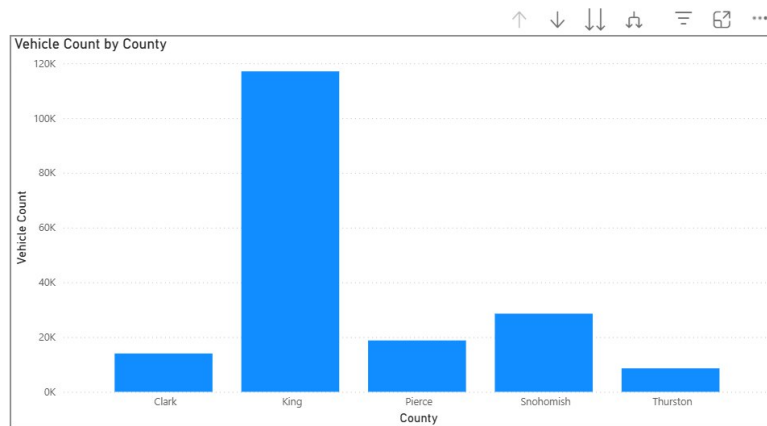


Figure 6: Top 5 Counties with the Highest Number of Vehicles

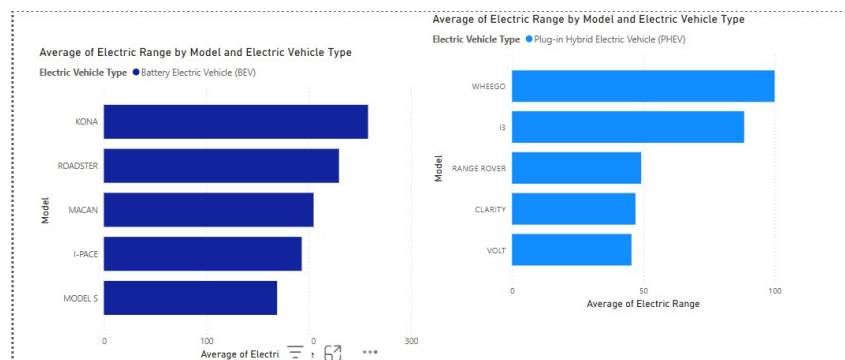


Figure 7: Top 5 Vehicles by Model and Electric Type